



Sylow intersections and Frobenius ratios

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Abstract. Let G be a finite group and p a prime dividing its order $|G|$, with p -part $|G|_p$, and let G_p denote the set of all p -elements in G . A well known theorem of Frobenius tells us that $f_p(G) = |G_p|/|G|_p$ is an integer. As G_p is the union of the Sylow p -subgroups of G , this *Frobenius ratio* $f_p(G)$ evidently depends on the number $s_p(G) = |\text{Syl}_p(G)|$ of Sylow p -subgroups of G and on *Sylow intersections*. One knows that $s_p(G) = 1 + kp$ and $f_p(G) = 1 + \ell(p-1)$ for nonnegative integers k, ℓ , and that $f_p(G) < s_p(G)$ unless G has a normal Sylow p -subgroup. In order to get lower bounds for $f_p(G)$ we, study the permutation character $\pi = \pi_p(G)$ of G in its transitive action on $\text{Syl}_p(G)$ via conjugation (Sylow character). We will get, in particular, that $f_p(G) \geq s_p(G)/r_p(G)$ where $r_p(G)$ denotes the number of P -orbits on $\text{Syl}_p(G)$ for any fixed $P \in \text{Syl}_p(G)$. One can have $\ell \geq k \geq 1$ only when P is irredundant for G_p , that is, when P is not contained in the union of the $Q \neq P$ in $\text{Syl}_p(G)$ and so $\hat{P} = \bigcup_{Q \neq P} (P \cap Q)$ a proper subset of P . We prove that $\ell \geq k$ when $|\hat{P}| \leq |P|/p$.

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1. Introduction. Throughout G will be a finite group and p a prime dividing its order $|G|$. By Sylow, the set $\text{Syl}_p(G)$ of Sylow p -subgroups of G is nonempty, and G acts transitively on it via conjugation. Let $\pi = \pi_p(G)$ denote the associated permutation character of G (*Sylow character*).

Fix $P \in \text{Syl}_p(G)$. Then $|P| = |G|_p$ equals the p -part of $|G|$ and $\text{Syl}_p(G) = P^G = \{P^g : g \in G\}$ (where $P^g = g^{-1}Pg$). Each P -orbit on $\text{Syl}_p(G)$ distinct from $\{P\}$ has size a proper power of p . Hence the *Sylow p -number* $s_p(G) = |\text{Syl}_p(G)|$ has the form $s_p(G) = 1 + kp$ for some integer $k \geq 0$. The set $G_p = \bigcup_{g \in G} P^g$ of all p -elements in G can be described as $G_p = \{g \in G : g^{|P|} = 1\}$. Hence, by a theorem of Frobenius [2], $f_p(G) = |G_p|/|P|$ is an integer, called

the *Frobenius p -ratio* of G . The identity

$$f_p(G) = \frac{1}{|P|} \sum_{x \in P} \frac{s_p(G)}{\pi(x)}$$

is basic for our purposes (see Lemma 3 below). Obviously $s_p(G)$ and $f_p(G)$ remain unchanged when replacing G by $G/O_p(G)$ where $O_p(G) = \bigcap_{g \in G} P^g$ is the greatest normal p -subgroup of G . We have $P = O_p(G)$ if and only $s_p(G) = 1$ resp. $f_p(G) = 1$.

We say that the Sylow p -subgroups of G are *irredundant* (for G_p) provided none can be omitted in order to get G_p as their union. Of course this is true if and only if the given $P \in \text{Syl}_p(G)$ cannot be omitted. Thus we have irredundancy in this sense precisely when $\hat{P} = \bigcup_{Q \in \text{Syl}_p(G) \setminus \{P\}} (P \cap Q)$ is a proper subset of P . So P is a T.I. subset of G just when $|\hat{P}| = 1$ and, as a rule, $f_p(G)$ is large when $|\hat{P}|$ is small. We call P *strongly irredundant* provided $|\hat{P}| \leq |P|/p$.

Theorem 1. *Suppose $P \in \text{Syl}_p(G)$ is not normal in G , so that the Sylow number $s_p(G) = 1 + kp$ with $k \geq 1$. Then the Frobenius ratio $f_p(G) = 1 + \ell(p-1)$ for some positive integer $\ell < \frac{kp}{p-1}$. We have $\ell \geq k$ when P is strongly irredundant, where $\ell > k$, unless $|P/O_p(G)| = p$. Conversely, if $\ell \geq k$, then P is irredundant.*

Thus $f_p(G) \geq p$ (as $\ell \geq 1$). We call k and ℓ the *parameters* of $s_p(G)$ and $f_p(G)$, respectively. If $|G/O_p(G)|_p = p^a$, then $\ell \leq \frac{kp}{p-1}(1 - \frac{1}{p^a})$, with equality precisely when $P/O_p(G)$ is a T.I. subset of $G/O_p(G)$. Observe that P is strongly irredundant when $P/O_p(G)$ is cyclic or, more generally, when $\hat{P} = \bigcup_{x \in P} (P \cap Q)^x$ for some $Q \in \text{Syl}_p(G)$. To see this, use that cyclic p -groups are uniserial, and that the normal closure of a proper subgroup of a finite p -group is a proper subgroup.

There exist groups G where $s_p(G) > |G_p|$, and then at least one Sylow p -subgroup of G is redundant for G_p , and we have $\ell < k$ (see the example in Sect. 2). One has $(1 + k(p-1))^p > (1 + kp)^{p-1}$ by binomial expansion. Hence $f_p(G) > s_p(G)^{\frac{p-1}{p}}$ when $\ell \geq k \geq 1$, and it is conjectured that this inequality holds in general. Indeed, Gheri [3] has confirmed this in the case where G is p -solvable, and has reduced this problem to almost simple groups. The authors have verified the conjecture for various such groups of small orders. It holds when G is a group of Lie type of characteristic p ; use that then $O_p(G) = 1$ and that $f_p(G) = |G|_p$ by a celebrated theorem of Steinberg [9].

Let $r_p(G)$ denote the number of P -orbits on $\text{Syl}_p(G)$, which evidently does not depend on the chosen $P \in \text{Syl}_p(G)$. If $s_p(G) = 1 + kp$ with $k \geq 1$, then $2 \leq r_p(G) \leq 1 + k$. We have $r_p(G) = 2$ if and only if P is transitive on the (nonempty) set $\text{Syl}_p(G) \setminus \{P\}$, which happens when G is a Zassenhaus group of suitable degree. Also, $r_p(G) = 1 + k$ if and only if each P -orbit on $\text{Syl}_p(G) \setminus \{P\}$ has size p .

Theorem 2. Let $s_p(G) = 1 + kp$ with $k \geq 1$. Then $f_p(G) > \frac{s_p(G)}{r_p(G)}$. The Sylow p -subgroup P of G is strongly irredundant for G_p in the extreme cases $r_p(G) = 2$ and $r_p(G) = 1 + k$.

In particular, we have strong irredundancy when $k \leq p$ since then $r_p(G) = 2$ or $r_p(G) = 1 + k$.

Theorem 3. Let $s_p(G) = 1 + kp$ with $k \geq 2$. Then $f_p(G) = 1 + \ell(p - 1)$ with $\ell \geq 2$, and the Sylow p -subgroup P of G is irredundant for G_p if $s_p(G) \leq p(f_p(G) + 2)$.

Thus $f_p(G) \geq 2p - 1$ when $s_p(G) \geq 1 + 2p$, and we certainly have irredundancy when $s_p(G) \leq p(2p + 1)$.

2. The Sylow character. We keep the assumptions and notations as introduced above. Fix $P \in \text{Syl}_p(G)$. We assume that P is not normal in G so that the Sylow number $s_p(G) = 1 + kp$ with $k \geq 1$. Then the Frobenius ratio $f_p(G) > 1$ as well. Recall that the Sylow character $\pi = \pi_p(G)$ for the prime p is the permutation character of G in its action on $\text{Syl}_p(G) = P^G$ (counting fixed points). For $x \in G$, let $x^G = \{x^g : g \in G\}$ denote the conjugacy class of x in G (so that $|x^G| = |G : C_G(x)|$).

Lemma 1. Let $N = N_G(P)$ be the normalizer of P in G , and let $\bigsqcup_{i=1}^r Ng_iP$ be the decomposition of G into double cosets mod (N, P) , say with $g_1 = 1$. Then $r = r_p(G)$ is the number of P -orbits on $\text{Syl}_p(G)$ and $s_p(G) = |G : N| = 1 + \sum_{i=2}^r |P : (P \cap P^{g_i})|$. In fact, the assignments $P^g \mapsto Ng$ for $g \in G$ define an isomorphism from the G -set $\text{Syl}_p(G)$ onto the G -set $G : N = \{Ng : g \in G\}$ (with G acting via multiplication from the right), and the P -sets $P : (P \cap P^{g_i})$, $i > 1$, may be identified with the P -orbits on $\text{Syl}_p(G) \setminus \{P\}$.

Proof. It is well known (and easily verified) that P^G is G -isomorphic to $G : N$ (see Aufg. 23 on page 32 in Huppert [5]). In particular, π is afforded by the G -set $G : N$, and is induced from the 1-character of N . So the remainder is nothing but Mackey decomposition (see Satz V.16.9 in [5]). The intersections $P \cap P^{g_i} = P \cap N^{g_i}$ are point stabilizers for the corresponding P -orbits, and are determined up to P -conjugacy. $\square$

Note that $N = N_G(P)$ is an abnormal subgroup of G [5, VI.11.18]; since $N_G(N) = N$, one also knows that $Ng \mapsto N^g$ for $g \in G$ defines an isomorphism from $G : N$ onto N^G . Of course $\text{Ker}(\pi) = \bigcap_{g \in G} N^g$, and $\pi(x) \geq 1$ if and only if $x \in \bigcup_{g \in G} N^g$. Also $O_p(G)$ is the kernel of P in its action on $\text{Syl}_p(G)$, and $\text{Ker}(\pi)/O_p(G)$ is a p' -group.

It is an interesting open question whether $s_p(G) = |G : N_G(P)|$ can be read off from the character table of G (Navarro). Can one then obtain in addition π as sum of the irreducible characters of G ?

Lemma 2. Let $x \in P$ and $N = N_G(P)$. We have $\pi(x) = s_p(G) \cdot \frac{|x^G \cap N|}{|x^G|}$ and $x^G \cap N = x^G \cap P$. Moreover $\pi(x)$ is the number of Sylow p -subgroups of G containing x (including P), and $\pi(x) = 1 + k_x p$ for some nonnegative integer

$k_x \leq k$. If H is a normal subgroup of G , then $s_p(G) = s_p(PH) \cdot s_p(G/H)$ and $\pi(x) = \pi_p(PH)(x) \cdot \pi_p(G/H)(Hx)$.

Proof. The first statement is well known (see Aufg. 24 on page 32 of [5]). The p -element $x \in P$ normalizes some Sylow p -subgroup Q of G if and only if $x \in Q$ (as Q is a p -subgroup of G of largest possible order, and $Q\langle x \rangle$ is a p -subgroup of G). Moreover $x^G \cap N = x^G \cap P$. We also have $\pi(x) \equiv 1 \pmod{p}$ since each $\langle x \rangle$ -orbit on $\text{Syl}_p(G)$ has size a power of p , and since $|\text{Syl}_p(G)| = s_p(G) = 1 + kp$. The last statement is due to Gheri [3, Lemma 3.6]. Use here that PH/H is a Sylow p -subgroup of G/H with normalizer NH/H , and consider the partition of the set of Sylow p -subgroups of G containing x (having cardinality $\pi(x)$) where P and Q belong to the same part if $PH = QH$. Since the Sylow p -subgroups of G/H are conjugate, there are $\pi_p(G/H)(Hx)$ parts with each part having cardinality $\pi_p(PH)(x)$. $\square$

Lemma 3. We have $f_p(G) = \frac{1}{|P|} \sum_{x \in P} \frac{s_p(G)}{\pi(x)}$.

Proof. Obviously $G_p = \bigcup_{x \in P} x^G$. In this union, for any $x \in P$, the conjugacy class x^G occurs with multiplicity $|x^G \cap P|$. Thus $|G_p| = \sum_{x \in P} \frac{|x^G|}{|x^G \cap P|}$. Apply the preceding lemma. $\square$

In view of Lemma 3, and adopting an idea of Gheri [3], we define

$$f_p^*(G) = s_p(G) / \left(\prod_{x \in P} \pi(x) \right)^{\frac{1}{|P|}}.$$

Of course this again is independent of the chosen particular Sylow p -subgroup of G . By the arithmetic–geometric mean inequality, we have $f_p(G) \geq f_p^*(G)$, and this inequality is proper since $s_p(G) > 1$, (and $\pi(x) \neq \pi(1)$ for some $x \in P$). Clearly $s_p(G)$, $r_p(G)$, $f_p(G)$, and $f_p^*(G)$ remain unchanged when replacing G by $G/O_p(G)$ (or by $O_{p'}(G)$). Gheri [3] conjectured that always $f_p^*(G) \geq s_p(G)^{\frac{p-1}{p}}$, and confirmed this when G is p -solvable. By Lemma 2, we know that $s_p(G) = s_p(PH) \cdot s_p(G/H)$ and $f_p^*(G) = f_p^*(PH) \cdot f_p^*(G/H)$ for any normal subgroup H of G . Arguing by induction on $|G|$, one thus is readily reduced to the case where $G = PH$ for some minimal normal subgroup H of G . Here H will be a p' -group when G is p -solvable, whence G p -nilpotent. This will be treated in Lemma 6 below.

Lemma 4. Let $\hat{P} = \bigcup_{Q \in \text{Syl}_p(G) \setminus \{P\}} (P \cap Q)$ be as introduced above. Then $\hat{P}$ is the subset of P consisting just of those $x \in P$ for which $\pi(x) > 1$; in other words, $\pi(x) = 1$ if and only if $x \in P \setminus \hat{P}$.

Proof. This is immediate from Lemma 2. $\square$

Lemma 5. Suppose that $O_p(G) = 1$, and let $1 \neq x \in P$. Then $\pi(x) \leq \frac{s_p(G)}{1+p}$ and hence $k_x \leq \frac{k-1}{1+p}$ (where $s_p(G) = 1 + kp$ and $\pi(x) = 1 + k_x p$ are as above).

Proof. We use an idea given in the proof of Proposition 2.1 in [4]. This proposition might be incorrect as it stands, its proof being not conclusive as the

sentence following statement (*) can only be substantiated when the p -element considered has order p .

We have $\pi(x) \leq \pi(y)$ if $\langle y \rangle \subseteq \langle x \rangle$. So we may let x have order p . We assert that there is $g \in G$ such that the sets of Sylow p -subgroups of G containing x and x^{gx^i} for $i = 0, \dots, p-1$ are pairwise disjoint. Then we have $1 + p$ such sets, each of cardinality $\pi(x)$ as π is a class function on G . Hence then $\pi(x)(1 + p) \leq s_p(G)$, as desired.

Assume the assertion is false. Then, for every $g \in G$, there are indices $i = i(g) \neq j(g) = j$ such that $\langle x, x^g \rangle = \langle x, x^{gx^i} \rangle^{x^{-i}}$ or $\langle x, x^{gx^{j-i}g^{-1}} \rangle = \langle x^{gx^i}, x^{gx^j} \rangle^{x^{-i}g^{-1}}$ are p -subgroups of G . In the latter case, $j - i$ is not divisible by p . Application of a variant of the Baer–Wielandt theorem now gives us that $\langle x \rangle$ is subnormal in G , whence $x \in O_p(G)$ in contrast to our hypothesis.

The variant we mean is as follows. Let $A = \langle x \rangle$ be a cyclic p -subgroup of G . Suppose that for every $g \in G$, the join $\langle x, x^{gyg^{-1}} \rangle$ is a p -group where either $y = g$ or $y = x^i$ for some integer $i = i(g)$ not divisible by p . Then we claim that A is subnormal in G . Arguing by induction on $|G|$, and assuming that the claim is false, by Wielandt's *Zipper Lemma* (originally proved in [10, Satz 5]), A is contained (and subnormal) in a unique maximal subgroup $M = N_G(M)$ of G . For any $g \in G$, this forces that either $A^g \subseteq M$ ($y = g$), $A \subseteq M^{g^{-1}}$, and $g^{-1} \in M$, or that $A^{gx^i g^{-1}} \subseteq M$, $(gx^i g^{-1})^{-1} \in M$, $A = \langle x^{-i} \rangle \subseteq M^g$, and $g \in M$. Thus, in each case, we get the contradiction $G \subseteq M$. $\square$

Lemma 6. *Suppose that $G = PH$ is p -nilpotent, with normal p -complement H . Then $s_p(G) = |H : C_H(P)|$ and $\pi(x) = |C_H(x) : C_H(P)|$ for any $x \in P$. Moreover*

$$\left(\prod_{x \in P} \pi(x) \right)^{\frac{1}{|P|}} \leq s_p(G)^{\frac{1}{p}},$$

where we have equality if and only if $C_H(x^p) = H$ for every $x \in P$. In particular, equality holds if P has exponent p , and then $f_p^*(G) = s_p(G)^{\frac{p-1}{p}}$.

Proof. We have $|G : N_G(P)| = |H : C_H(P)|$ and $x^G \cap P = x^P$ for $x \in P$ [5, Satz IV.4.9]. From [5, Satz I.8.6], it follows that $C_G(x)H = C_P(x)H$, whence $|G : C_G(x)|$ equals $|P : C_P(x)| \cdot |H : C_H(x)|$. Use finally that $|C_H(P)|^{|P|} = \left(\prod_{x \in P} \frac{|C_H(x)|}{|C_H(x^p)|^{\frac{1}{p}}} \right)^{\frac{p}{p-1}}$ by Theorem A in Navarro–Rizo [8]. $\square$

Example. For odd p the first example where a Sylow p -subgroup is redundant for G_p has been given by Derek Holt, namely for $G = \text{PSL}_3(7)$ and $p = 3$. This can be found in a post given by Mikko Korhonen in MathematicsStackExchange (2013). There one also finds the following example by Jack Schmidt (see also Example 2.5 in [3]):

Let P be elementary abelian of order p^a ($a \geq 2$). Let $q \equiv 1 \pmod{p}$ be a prime and let H be an elementary abelian q -group of rank $\frac{p^a-1}{p-1}$, which is the number of maximal subgroups M_i of P . There is a faithful action of P on H such that $H/C_H(M_i)$ has order q for each i (and $\bigcap_i C_H(M_i) = 1$). The corresponding semidirect product $G = H \cdot P$ is a p -nilpotent group satisfying

$C_H(P) = 1$ and $|C_H(x)| = q^{\frac{p^a-1}{p-1}}$ for any $x \neq 1$ in P , because there are just $\frac{p^a-1}{p-1}$ maximal subgroups of P containing x . Hence $s_p(G) = |H|$ and $\pi(x) = q^{\frac{p^a-1}{p-1}}$ by Lemma 6. We infer that P is redundant for G_p and that $f_p^*(G) = s_p(G)^{\frac{p-1}{p}}$. Moreover $|G_p| = 1 + (p^a - 1)q^{p^{a-1}}$. We have $q^{\frac{p^a-1}{p-1}} > 1 + (p^a - 1)q^{p^{a-1}}$ provided q is chosen large enough compared with p^a (which is possible by Dirichlet's prime number theorem). In this case $s_p(G) > |G_p|$.

3. Proof of Theorem 1. Fix $P \in \text{Syl}_p(G)$, and let $O_p(G) = 1$ (without loss). Write $|P| = p^a$ and $s_p(G) = 1 + kp$ ($a \geq 1, k \geq 1$). Let also $\pi = \pi_p(G)$. Two nontrivial p -elements x, y of G are said to be *related* provided $\langle x \rangle = \langle y \rangle$. This is an equivalence relation on $G_p^\# = G_p \setminus \{1\}$, each class having cardinality divisible by $p - 1$. Thus $|G_p^\#| = p^a f_p(G) - 1 = (p^a - 1)f_p(G) + (f_p(G) - 1)$ is divisible by $p - 1$. Since $p - 1$ is a divisor of $p^a - 1$, we get that $f_p(G) - 1 = \ell(p - 1)$ for some integer ℓ , as required. Clearly $\ell \geq 1$ (as $k \geq 1$). From Lemmas 3 and 2, it follows that $f_p(G) < s_p(G)$, whence $\ell < \frac{kp}{p-1}$.

Let $\widehat{P} = \{x \in P : \pi(x) > 1\}$ (Lemma 4). If P is a T.I. subset of $G = G/O_p(G)$, then $|\widehat{P}| = 1$ and $f_p(G) = \frac{1}{p^a}(p^a(1 + kp) - kp) = 1 + kp(1 - \frac{1}{p^a}) = s_p(G) - \frac{kp}{p^a}$ takes its greatest value (in terms of $s_p(G)$ and $|P|$). Here p^{a-1} is a divisor of k (Frobenius). Let $|\widehat{P}| > 1$ in what follows, which forces that $a \geq 2$. Using that $\pi(x) < s_p(G)$ for $1 \neq x \in \widehat{P}$ and $\pi(x) = 1$ for $x \in P \setminus \widehat{P}$ from Lemma 3, we get that

$$f_p(G) > \frac{1}{|P|}(|\widehat{P}| + (|P| - |\widehat{P}|)(1 + kp)).$$

If P is strongly irredundant, then $|\widehat{P}| \leq |P|/p = p^{a-1}$ (by definition), and we obtain that $f_p(G) > 1 + kp - \frac{kp|\widehat{P}|}{|P|} \geq 1 + k(p - 1)$.

Assume that $f_p(G) \geq 1 + k(p - 1)$ but that P is redundant. Then $\pi(x) \geq 1 + p$ for each $x \in P$ by Lemmas 4 and 2, whence by Lemma 3,

$$1 + k(p - 1) \leq f_p(G) \leq \frac{1 + kp}{1 + p}.$$

It follows that $k(p^2 - p - 1) \leq -p$, which is impossible as $k > 0$ and $p^2 - p - 1 > 0$.

Remark. Arguing as above, one gets that $|\widehat{P}| = 1 + m(p - 1)$ for some integer $m \geq 0$ (even $|\widehat{P}|/|\widehat{P}|_p$ is of this form). This could be helpful in order to check whether P is strongly irredundant. Of course there are groups G where a Sylow p -subgroup is irredundant but not strongly irredundant, and where the parameters $\ell < k$. (For $p = 2$, such an example is provided by the symmetric group $G = S_5$, where $s_2(G) = 15$ ($k = 7$) and $f_2(G) = 7$ ($\ell = 6$), and where $\pi(x) = 1$ for any 4-cycle $x \in G$.)

4. Proof of Theorem 2. Without loss let again $O_p(G) = 1$, and let $s_p(G) = 1 + kp$ with $k \geq 1$. Fix $P \in \text{Syl}_p(G)$ and let $\pi = \pi_p(G)$ be the Sylow character.

By the Cauchy–Frobenius–Burnside fixed point formula [5, Satz V.13.4] $r_p(G) = \frac{1}{|P|} \sum_{x \in P} \pi(x)$, which is greater than $(\prod_{x \in P} \pi(x))^{\frac{1}{|P|}}$ by the arithmetic–geometric mean inequality. Consequently

$$f_p(G) > f_p^*(G) > \frac{s_p(G)}{r_p(G)},$$

as desired. Clearly $2 \leq r_p(G) \leq 1 + k$, and $r_p(G) = 2$ if and only if P is transitive on $\text{Syl}_p(G) \setminus \{P\}$, in which case $\hat{P} = \bigcup_{x \in P} (P \cap Q)^x$ for any $Q \neq P$ in $\text{Syl}_p(G)$.

Suppose that $r_p(G) = 1 + k$. Then each P -orbit on $\text{Syl}_p(G)$ different from $\{P\}$ has size p . It follows that $P = P/O_p(G)$ is elementary abelian (as it acts faithfully on $\text{Syl}_p(G) \setminus \{P\}$). By Brodkey [1] there is $Q \in \text{Syl}_p(G)$ such that $P \cap Q = 1$. But $P \cap Q$ is a point stabilizer for the P -action on $\text{Syl}_p(G)$ and $|P : (P \cap Q)| = p$ is the size of a P -orbit. Thus $|P| = p$ and, therefore, P is a T.I. subset of G . It follows that $|G_p| = (1 + kp)p - kp$ and $f_p(G) = 1 + k(p - 1)$. (In general there is no $Q \in \text{Syl}_p(G)$ such that $P \cap Q = O_p(G)$; e.g. see Ito [6].)

Remark. Let $D = P \cap Q$ be a Sylow intersection of largest order ($P \neq Q \in \text{Syl}_p(G)$). Let $|P : D| = p^b$ and $|N_P(D) : D| = p^c$ ($1 \leq c \leq b$). Then the contribution to $f_p(G)$ resulting from $P \cup \bigcup_{x \in N_P(D)} Q^x$ is just $1 + p^c - p^{c-b}$, because the $1 + p^c$ Sylow subgroups involved intersect pairwise in D . Hence $f_p(G) > p$ except possibly when $b = c = 1$.

5. Proof of Theorem 3. Let again $O_p(G) = 1$ (without loss), and let $s_p(G) = 1 + kp$ with $k \geq 2$. Fix also $P \in \text{Syl}_p(G)$, and let $|P| = p^a$. By Theorem 1, we know that $f_p(G) = 1 + \ell(p - 1)$ for some integer $\ell \geq 1$. As usual, we let $\pi = \pi_p(G)$.

We first show that the parameter $\ell \geq 2$. Otherwise $\ell = 1$ and $f_p(G) = p$, and we must have $a \geq 2$ (since otherwise P is a T.I. subset of G). From Lemma 5 we get that $\pi(x) \leq \frac{s_p(G)}{1+p}$ for each $x \neq 1$ in P . Applying Lemma 3 we obtain that

$$p = f_p(G) \geq \frac{1}{p^a} + \frac{p^a - 1}{p^a} (1 + p) = 1 + p - \frac{1}{p^{a-1}} > p,$$

which is the desired contradiction. In particular, $f_p(G) \geq 1 + 2(p - 1) = 2p - 1$.

(A different approach can be given using the remark following Theorem 2, which in turn is in the spirit of [7, p. 80].)

Now let $s_p(G) \leq p(f_p(G) + 2)$ or, equivalently, let $k < f_p(G) + 2$. We assert that then P is irredundant. Assume the assertion is false. It follows from Theorem 2 that we must have $k \geq 1 + p$, for otherwise P were even strongly irredundant. From Lemmas 4 and 2, we get that $\pi(x) \geq 1 + p$, for each $x \in P$. Using that $\pi(1) = 1 + kp > 1 + p$ therefore Lemma 3 shows that

$$2p - 1 \leq f_p(G) < \frac{1 + kp}{1 + p} = k - \frac{k - 1}{1 + p} \leq k - \frac{p}{1 + p}.$$

Consequently $k > 2p - 1 + \frac{p}{1+p}$, whence $k \geq 2p \geq p + 2$. But this implies that $f_p(G) < k - \frac{k-1}{1+p} \leq k - 1$ and so $k \geq f_p(G) + 2$, against our hypothesis.

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